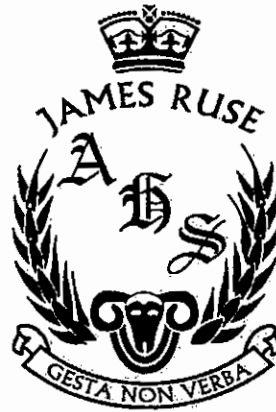


Name:	
Class:	



YEAR 12

ASSESSMENT TASK 2

TERM 1, 2017

MATHEMATICS

EXTENSION 2

*Time Allowed – 120 Minutes
(Plus 5 minutes Reading Time)*

General Instructions:

- All questions may be attempted
- All questions are of equal value
- Reference Sheet will be supplied
- Department of Education approved calculators and templates are permitted
- In every Question, show all relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless or badly arranged work

**Question 1-5 are to be completed on multiple choice sheet then Question 6-10 to be completed on separate sheets of paper and handed in in separate bundles.
Each question must show your Candidate Number (in the top right hand corner).**

QUESTION ONE

The value of $\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx$ is which of the following?

- (A) $e - 1$
- (B) $e + 1$
- (C) e
- (D) 0

QUESTION TWO

Which of the following is the equation of the hyperbola with foci $(6, 5)$ and $(-4, 5)$, and eccentricity $\frac{5}{4}$?

- (A) $\frac{(x-1)^2}{16} - \frac{(y-5)^2}{9} = 1$
- (B) $\frac{(x-1)^2}{16} - \frac{(y-5)^2}{9} = -1$
- (C) $\frac{x^2}{16} - \frac{y^2}{9} = 1$
- (D) $\frac{x^2}{16} - \frac{y^2}{9} = -1$

QUESTION THREE

The zeros of the polynomial $z^4 + az^3 + bz^2 + cz + d$ are all complex numbers that lie on the unit circle. The coefficients a, b, c and d are real numbers. What is the sum of the reciprocals of the roots of the polynomial?

- (A) a
- (B) $-a$
- (C) c
- (D) $-c$

Examination continues on next page ...

QUESTION FOUR

Suppose that n is a positive integer and that $z + \frac{1}{z} = 2 \cos \theta$, for $0 < \theta < \pi$. What is the value of $z^n + \frac{1}{z^n}$?

- (A) $2(\cos \theta)^n$
- (B) $2^n \cos n\theta$
- (C) $2(\cos \theta)^n$
- (D) $2 \cos n\theta$

QUESTION FIVE

The function P satisfies the equation $P(2+x) = P(2-x)$ for all real numbers x . Also, $P(x) = 0$ has four distinct real roots. What is the sum of these roots?

- (A) 8
- (B) 6
- (C) 4
- (D) 2

QUESTION SIX (Start a new answer booklet.)

Marks

- (a) Evaluate $\int_0^1 \frac{x^2 + x}{\sqrt{x+1}} dx$. 2
- (b) Find $\int \tan^3 \theta d\theta$. 2
- (c) Use the substitution $u = \sqrt{1-x}$ to find $\int_0^1 x^2 \sqrt{1-x} dx$. 3
- (d) Define the integral J_n as $J_n = \int (\log_e x)^n dx$. 4
- (i) Use integration to show that $J_n = x(\log_e x)^n - nJ_{n-1}$.
- (ii) Hence show that $\int_1^e (\log_e x)^2 dx = e - 2$.
- (e) Evaluate $\int \frac{5x^2 - 4x + 5}{(x-1)(x^2+2)} dx$ by use of partial fractions. 3
- (f) Using $t = \tan \frac{x}{2}$, evaluate $\int \frac{1}{1 + \sin x} dx$. Leave your answer in terms of $\tan \frac{x}{2}$. 2

Examination continues on next page ...

QUESTION SEVEN (Start a new answer booklet.)

Marks

- (a) If α, β, γ are the roots of the equation $3x^3 - 5x^2 - 4x + 3 = 0$, find the cubic equation having roots $\alpha - 1, \beta - 1, \gamma - 1$. 3
- (b) The cubic equation $x^3 + px + q = 0$ has three non-zero roots α, β, γ . 3
Find, in terms of the constants p and q , the values of
- (i) $\alpha^2 + \beta^2 + \gamma^2$
- (ii) $\alpha^3 + \beta^3 + \gamma^3$
- (c) For the ellipse with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$:
- (i) Sketch the ellipse, showing the foci S and S' , and the directrices. 4
- (ii) Prove that the tangent to the ellipse at the point $P(4 \cos \theta, 3 \sin \theta)$ has the equation $\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1$. 3
- (iii) If the ellipse meets the y -axis at B and B' , and the tangents at B and B' meet the tangent at P at the points Q and Q' , prove that $BQ \cdot B'Q' = 16$. 3

Examination continues on next page ...

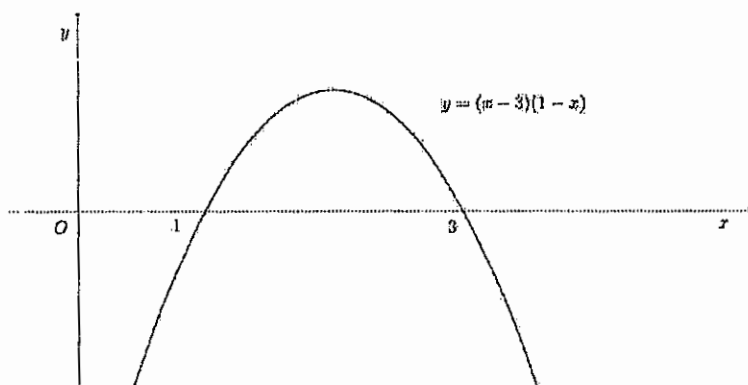
QUESTION EIGHT (Start a new answer booklet.)

Marks

- (a) (i) Express $\sqrt{3} + i$ in modulus-argument form. 2
- (ii) Hence simplify $(\sqrt{3} + i)^{12}$ using de Moivre's theorem. 2
- (b) The triangle OPQ lies in the complex plane such that O is the origin, $OP \perp PQ$, $OP = \frac{2}{\sqrt{3}}$ and $PQ = \frac{11}{2\sqrt{3}}$. Let X be a point on PQ such that $PX = 1$ and let z be the complex number representing X . 3
- By considering z^3 , show that $\angle XOP = \frac{1}{3}\angle QOP$.
- (c) (i) Let z be a complex number such that $z \neq 0$. Show that $\frac{z}{\bar{z}} - \frac{\bar{z}}{z}$ is purely imaginary. 2
- (ii) Hence sketch the locus of $\frac{z}{\bar{z}} - \frac{\bar{z}}{z}$. (Note, you may wish to consider z in the form of $z = r(\cos \theta + i \sin \theta)$, where $r = |z|$ and $\theta = \arg(z)$.) 2
- (d) The polynomial $P(z) = z^4 - 4z^3 + 24z^2 - 40z + 100$ has two non-real double zeros. Let α be one of these zeros so that $P(\alpha) = 0$.
- (i) Show that $P(\bar{\alpha}) = 0$. 2
- (ii) Let $\alpha = x + iy$. By considering the relationship between the zeros of a polynomial and the coefficients of its terms, find x and y , and hence solve $P(z) = 0$. 3

QUESTION NINE (Start a new answer booklet.)

Marks



- (a) The diagram above shows the graph of $y = f(x)$ where $f(x) = (x-3)(1-x)$.

On separate diagrams, and without the use of calculus, sketch the following graphs, clearly indicating any significant features.

- | | |
|---------------------------|---|
| (i) $y = f(x) $ | 2 |
| (ii) $y = \frac{1}{f(x)}$ | 2 |
| (iii) $y = e^{f(x)}$ | 2 |
| (iv) $y^2 = f(x)$ | 2 |
- (b) The hyperbolae $H_1 : x^2 - \frac{y^2}{8} = 1$ and $H_2 : y^2 - x^2 = 1$ are known to intersect. 3

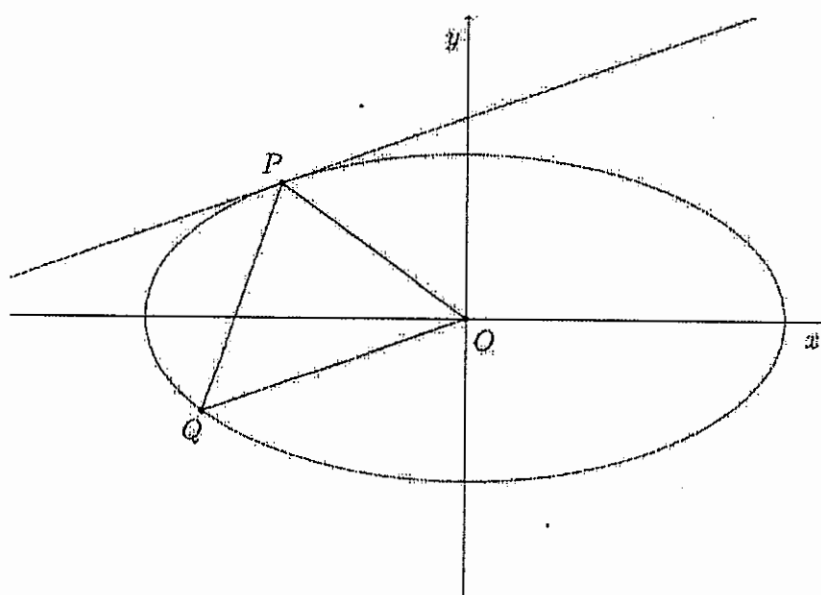
Let C be the family of curves

$$C : \left(x^2 - \frac{y^2}{8} - 1\right) + k(y^2 - x^2 - 1) = 0$$

that pass through the points of intersection of H_1 and H_2 , for k a real number.

For what values of k will the family of curves C be hyperbolic and with its foci on the x -axis?

Examination continues on next page ...



- (c) The diagram above is the graph of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Let $P = (a \cos \alpha, b \sin \alpha)$ and $Q(a \cos \beta, b \sin \beta)$ be two points on the ellipse such that OQ is parallel to the tangent at P , and where O is the origin.

- (i) By considering the gradient of the tangent at P and the gradient of OQ , 2
show that $\cot \alpha = -\tan \beta$.
- (ii) Hence show that the area of triangle OPQ is $\frac{|ab|}{2}$. 3

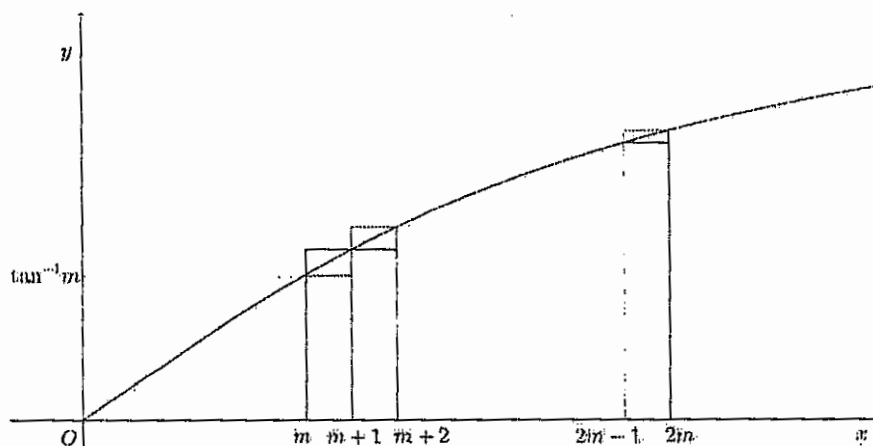
(Note, you may use the fact that for any two points $U = (x_1, y_1)$ and $V = (x_2, y_2)$ in the plane, $\frac{1}{2}|x_1y_2 - x_2y_1|$ gives the area of the triangle OUV .)

QUESTION TEN (Start a new answer booklet.)

Marks

- (a) The points P, Q , and R lie in the complex plane and represent the complex numbers z_1, z_2 , and z_3 respectively. Show that if triangle PQR is equilateral, then 4

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1.$$



- (b) The diagram above shows the graph of $y = \tan^{-1} x$ for $x \geq 0$. A construction of m upper, and m lower, rectangles has been made between $x = m$ and $x = 2m$, each of width one unit.

Let
$$S_m = \sum_{k=1}^m \tan^{-1}(m+k) = \tan^{-1}(m+1) + \tan^{-1}(m+2) + \cdots + \tan^{-1}(2m).$$

- (i) By considering the sums of the areas of the upper and lower rectangles, along with the integral $\int_m^{2m} \tan^{-1} x \, dx$, prove that 4

$$2m \tan^{-1}(2m) - m \tan^{-1} m + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right) < S_m$$

and that

$$S_m < (2m+2) \tan^{-1}(2m+2) - (m+1) \tan^{-1}(m+1) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right).$$

- (ii) Hence find 2

$$S_{1000} = \tan^{-1}(1001) + \tan^{-1}(1002) + \cdots + \tan^{-1}(2000)$$

to three significant figures.

Examination continues on next page ...

- (c) A polynomial $P(z)$ is said to be *reducible* over the integers $\mathbb{Z}$ if it can be written as a product of polynomials $Q(z)R(z) = P(z)$, where $Q(z)$ and $R(z)$ are both integer polynomials (that is, polynomials whose coefficients are all integers). If such a factorisation cannot be performed, we say that the polynomial is *irreducible*.

Define the polynomial $P(z)$ as

$$P(z) = z^n + z + p$$

where z is a complex number, n is a positive integer and $p > 2$ is a prime number.

- (i) Suppose that $P(z)$ may be factorised into a product of two integer polynomials as 1

$$P(z) = Q(z)R(z)$$

Show that either $Q(0) = \pm 1$ and $R(0) = \pm p$ or $R(0) = \pm 1$ and $Q(0) = \pm p$.

- (ii) Without loss of generality, take the case where $Q(0) = \pm 1$ and $R(0) = \pm p$. 2
Let the roots of $Q(z) = 0$ be $\alpha_1, \alpha_2, \dots, \alpha_k$ where k is some integer less than n .

Given that the product of the roots of a monic polynomial of degree k is given by

$$\alpha_1 \alpha_2 \cdots \alpha_k = (-1)^k a_0$$

where a_0 is the constant term of the polynomial, show that there is at least one root α_j of $Q(z)$ such that $|\alpha_j| \leq 1$.

- (iii) Hence, using the triangle inequality $|z + w| \leq |z| + |w|$ or otherwise, show 3
that $P(z)$ is *irreducible*.

END OF EXAMINATION

$$\int \frac{x^2+x}{\sqrt{x+1}} dx = \int_0^1 \frac{x(x+1)}{\sqrt{x+1}} dx$$

$$= \int_0^1 x \sqrt{x+1} dx$$

let $u = x+1$

$du = dx$

$x=0 \quad u=1$

$x=1 \quad u=2$

$$= \int_1^2 (u-1) u^{1/2} du$$

$$= \int_1^2 u^{3/2} - u^{1/2}$$

$$= \left[\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right]_1^2$$

$$= \frac{8\sqrt{2}}{5} - \frac{4\sqrt{2}}{3} - \frac{2}{5} + \frac{2}{3}$$

$$= \frac{4}{15} (1+\sqrt{2})$$

1

Students could use integration by parts but ~~was~~ were less successful

1

2nd mark was given for this

Answer should have been simplified

Suggested Solutions

Marks Awarded

Marker's Comments

$$\begin{aligned}
 (b) \int \tan^3 \theta &= \int \tan \theta \cdot \tan^2 \theta d\theta \\
 &= \int \tan \theta (\sec^2 \theta - 1) d\theta \\
 &= \int \tan \theta \sec^2 \theta d\theta - \int \frac{\sin \theta}{\cos \theta} d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} \tan^2 \theta + \log_e |\cos x| + C \\
 \text{or } &\frac{1}{2} \sec^2 \theta + \log_e |\cos x| + C.
 \end{aligned}$$

$$\begin{aligned}
 (c) \text{ let } u &= \sqrt{1-x} \\
 u^2 &= 1-x \\
 x &= 1-u^2 \\
 x^2 &= (1-u^2)^2
 \end{aligned}$$

$$dx = -2u du.$$

$$x=0 \quad u=1$$

$$x=1 \quad u=0$$

$$\int_0^1 x^2 \sqrt{1-x} dx = \int_1^0 (1-u^2)^2 u (-2u) du$$

Many students were careless and lost terms

Suggested Solutions

Marks Awarded

Marker's Comments

$$\begin{aligned}
 & 2 \int_0^1 u^2 - 2u^4 + u^6 du \\
 &= 2 \left[\frac{u^3}{3} - \frac{2u^5}{5} + \frac{u^7}{7} \right]_0^1 \\
 &= 2 \left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7} \right) \\
 &= \frac{16}{105}
 \end{aligned}$$

$x^2 \sqrt{1-x}$
 is positive
 so answer
 should be
 positive

depending
 on how many
 terms were
 lost CFE
 were not necessarily
 given.

$$\begin{aligned}
 (c) \quad I_n &= \int (\log x)^n dx \\
 &= \int (\log x)^n \cdot x \cdot \frac{1}{x} dx \\
 &= \int (\log x)^n \frac{d}{dx}(x) dx \\
 &= x(\log x)^n - \int x \times n(\log x)^{n-1} \cdot \frac{1}{x} dx \leftarrow 1 \\
 &= x(\log x)^n - n \int (\log x)^{n-1} dx \\
 &= x(\log x)^n - n I_{n-1}
 \end{aligned}$$

had to
 see this
 term smaller

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks Awarded

Marker's Comments

MATHEMATICS Extension 2: Question.....

4

Suggested Solutions

Marks Awarded

Marker's Comments

$$(ii) J_2 = \left[x(\ln x)^2 \right]_1^e - 2J_1$$

$$= e(\ln e)^2 - 1(\ln 1)^2 - 2J_1$$

$$J_1 = \left[x \ln x \right]_1^e - \int_1^e dx$$

$$= e \ln e - 1 \ln 1 - [x]_1^e$$

$$= e \ln e - (e - 1)$$

$$= 1$$

$$J_2 = \underline{e - 2}$$

$$(e) \frac{5x^2 - 4x + 5}{(x-1)(x^2+2)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+2}$$

~~$$\frac{A+B}{x-1} + \frac{C}{x^2+2}$$~~

$$5x^2 - 4x + 5 = A(x^2+2) + (Bx+C)(x-1)$$

$$\text{if } x=1$$

$$3A = 6$$

$$A = 2$$

Students made careless errors and skipped steps.

MATHEMATICS Extension 2: Question.....6

5

Suggested Solutions

Marks Awarded

Marker's Comments

if $x=0$

$$5 = 2A - C$$

$$5 = 4 - C$$

$$C = -1$$

if $x = -1$

$$14 = 3A + 2B - 2C$$

$$= 6 + 2B + 2$$

$$2B = 6$$

$$\underline{B = 3}$$

$$\therefore \int \frac{2}{x-1} dx + \int \frac{3x-1}{x^2+2} dx$$

$$= 2 \int \frac{dx}{x-1} + \frac{3}{2} \int \frac{2x dx}{x^2+2} - \int \frac{dx}{x^2+2}$$

$$= 2 \ln(x-1) + \frac{3}{2} \ln(x^2+2) - \frac{1}{\sqrt{2}} \tan^{-1} \frac{x}{\sqrt{2}} + C.$$

No carry forward error if did not have a $\tan^{-1}$ integral

MATHEMATICS Extension 2: Question.....

Suggested Solutions

Marks Awarded

Marker's Comments

$$I = \int \frac{1}{1 + \sin x} dx$$

$$t = \tan \frac{x}{2} \quad \sin x = \frac{2t}{1+t^2}$$

$$\frac{2}{1+t^2} dt = dx$$

$$I = \int \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{2}{1+t+t^2} dt$$

$$= \int \frac{2}{(t+1)^2} dt$$

$$= -2(t+1)^{-1} + C$$

$$= \frac{-2}{1 + \tan \frac{x}{2}} + C$$

1

1

Many students
lost mark
for not
substituting
back

MATHEMATICS Extension 2: Question...7...

Suggested Solutions

Marks

Marker's Comments

$$a) x = \alpha - 1$$

$$\therefore \alpha = x + 1$$

$\therefore$ equation will be:

$$3(x+1)^3 - 5(x+1)^2 - 4(x+1) + 3 = 0$$

$$3(x^3 + 3x^2 + 3x + 1) - 5(x^2 + 2x + 1) - 4x - 4 + 3 = 0$$

$$-4x - 4 + 3 = 0$$

$$3x^3 + 9x^2 + 9x + 3 - 5x^2 - 10x - 5 - 4x - 4 + 3 = 0$$

$$-4x - 4 + 3 = 0$$

$$3x^3 + 4x^2 - 5x - 3 = 0$$

$$b) x^3 + px + q = 0$$

$$\therefore \alpha + \beta + \gamma = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = p$$

$$\alpha\beta\gamma = -q$$

$$i) (\alpha + \beta + \gamma)^2 = \alpha^2 + \alpha\beta + \alpha\gamma + \alpha\beta + \beta^2 + \beta\gamma + \alpha\gamma + \beta\gamma + \gamma^2$$

$$= \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$\therefore \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= 0^2 - 2(p)$$

$$= -2p$$

$$ii) \alpha^3 + p\alpha + q = 0 \quad \text{--- (1)}$$

$$\beta^3 + p\beta + q = 0 \quad \text{--- (2)}$$

$$\gamma^3 + p\gamma + q = 0 \quad \text{--- (3)}$$

$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$\alpha^3 + \beta^3 + \gamma^3 + p(\alpha + \beta + \gamma) + 3q = 0$$

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -p(0) - 3q$$

$$= -3q$$

$$c) i) b^2 = a^2(1 - e^2)$$

$$9 = 16 - 16e^2$$

$$e^2 = \frac{7}{16}$$

$$e = \frac{\sqrt{7}}{4}$$

some students had incorrect sum & product of roots.

$$\text{i.e. } \alpha + \beta + \gamma = -p$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = q$$

students who used $(\alpha + \beta + \gamma)^3$ made more mistakes with the expansion.

MATHEMATICS Extension 2: Question...7...

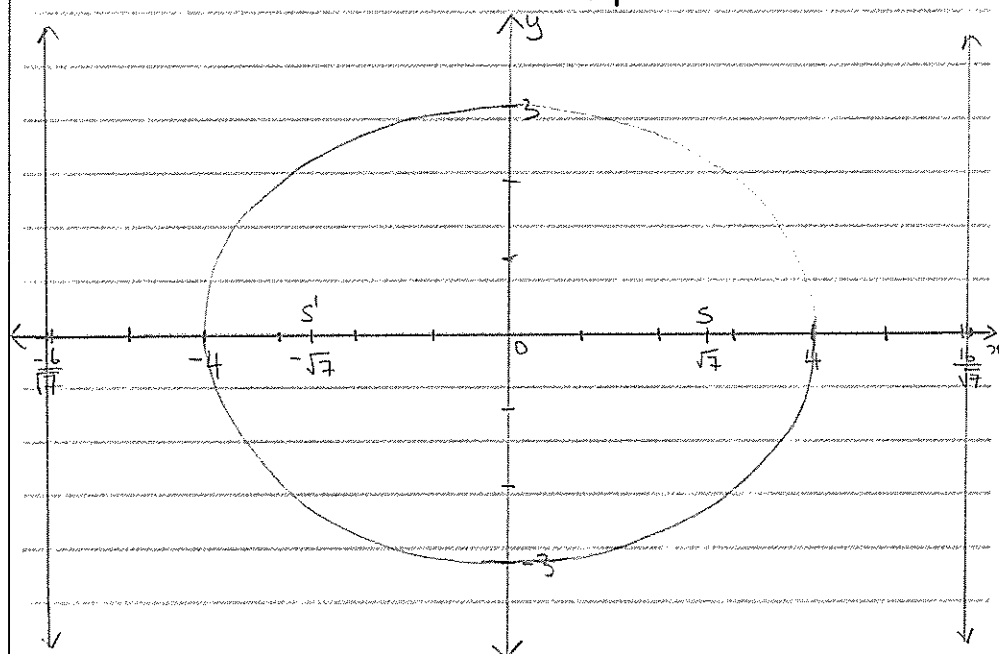
Suggested Solutions

Marks

Marker's Comments

$$S = (\pm\sqrt{7}, 0)$$

directrices: $x = \pm \frac{16}{\sqrt{7}}$



ii) $P(4\cos\theta, 3\sin\theta)$

$$x = 4\cos\theta$$

$$y = 3\sin\theta$$

$$\frac{dx}{d\theta} = -4\sin\theta$$

$$\frac{dy}{d\theta} = 3\cos\theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= -\frac{3\cos\theta}{4\sin\theta}$$

$\therefore$ equation of tangent is:

$$y - 3\sin\theta = -\frac{3\cos\theta}{4\sin\theta} (x - 4\cos\theta)$$

$$4y\sin\theta - 12\sin^2\theta = -3x\cos\theta + 12\cos^2\theta$$

$$3x\cos\theta + 4y\sin\theta = 12\sin^2\theta + 12\cos^2\theta$$

$$3x\cos\theta + 4y\sin\theta = 12(\sin^2\theta + \cos^2\theta)$$

$$3x\cos\theta + 4y\sin\theta = 12$$

$$\frac{x\cos\theta}{4} + \frac{y\sin\theta}{3} = 1$$

Need to show
- x & y intercepts
- focus
- directrices

Please make
sure you have
a graph that
is at least
 $\frac{1}{3}$ of the page.

differentiating
to find
gradient

need to
show how
you got the
12,
ie. $\sin^2\theta + \cos^2\theta = 1$

MATHEMATICS Extension 2: Question...7...

Suggested Solutions

Marks

Marker's Comments

iii) tangent at B: $y=3$
 $B': y=-3$

when $y=3$,

$$\frac{x \cos \theta}{4} + \frac{3 \sin \theta}{3} = 1$$

$$\frac{x \cos \theta}{4} = 1 - \sin \theta$$

$$x = \frac{4(1 - \sin \theta)}{\cos \theta}$$

$\therefore Q$ is $\left(\frac{4(1 - \sin \theta)}{\cos \theta}, 3 \right)$

$$BQ = \left| \frac{4(1 - \sin \theta)}{\cos \theta} \right|$$

when $y=-3$

$$\frac{x \cos \theta}{4} - \frac{3 \sin \theta}{3} = 1$$

$$\frac{x \cos \theta}{4} = 1 + \sin \theta$$

$$x = \frac{4(1 + \sin \theta)}{\cos \theta}$$

$\therefore Q'$ is $\left(\frac{4(1 + \sin \theta)}{\cos \theta}, -3 \right)$

$$B'Q' = \left| \frac{4(1 + \sin \theta)}{\cos \theta} \right|$$

$$\therefore BQ \cdot B'Q' = \left| \frac{4(1 - \sin \theta)}{\cos \theta} \right| \times \left| \frac{4(1 + \sin \theta)}{\cos \theta} \right|$$

$$= \left| \frac{16(1 - \sin^2 \theta)}{\cos^2 \theta} \right|$$

$$= \left| \frac{16 \cos^2 \theta}{\cos^2 \theta} \right|$$

$$= 16$$

1 for $Q \& Q'$

1 for distance of $BQ \& B'Q'$

1 for showing

$$BQ \cdot B'Q' = 16$$

need to see how

$$\frac{16(1 - \sin^2 \theta)}{\cos^2 \theta}$$

$$= 16 \frac{\cos^2 \theta}{\cos^2 \theta}$$

cannot just say $\frac{(1 - \sin^2 \theta)}{\cos^2 \theta}$

is 1, they are different terms!

Need to show why they are equal.

Mathematics Ext2 Question 8

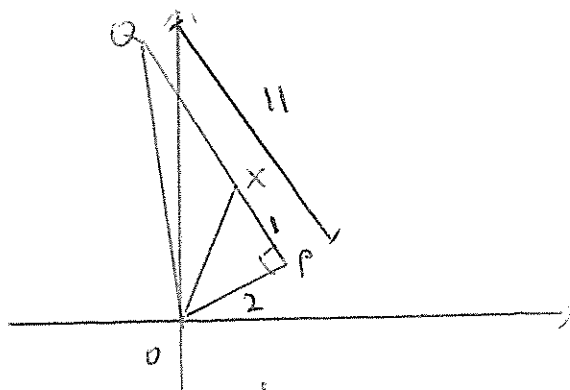
Suggested Solutions	Marks	Marker's Comments
a) i) let $z = \sqrt{3} + i$ $ z = \sqrt{(\sqrt{3})^2 + 1^2}$ $= 2$ $\text{Arg}(z) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$ $= \frac{\pi}{6}$ $\therefore z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$ <u>Note</u> : $z = 2\text{cis} \frac{\pi}{6}$ is insufficient!	— (1) — (1)	For either just the argument or the modulus
ii) $(\sqrt{3} + i)^{12} = 2^{12} \left(\cos \frac{12\pi}{6} + i \sin \frac{12\pi}{6} \right)$ (De Moivre's) $= 2^{12} (\cos 2\pi + i \sin 2\pi)$ $= 2^{12}$ $= 4096$	— (1) — (1)	

Suggested Solutions

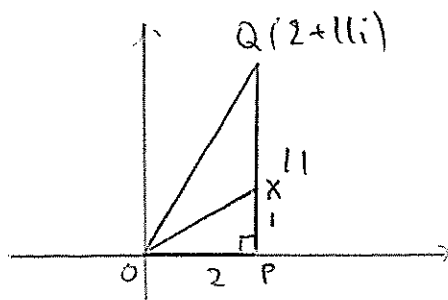
Marks

Marker's Comments

b)



↓ is congruent to



$$\begin{aligned}
 z^3 &= (2+i)^3 \\
 &= 2^3 + 3 \cdot 2^2 i + 3 \cdot 2 i^2 + i^3 \quad (\text{Binomial Expansion}) \\
 &= 8 + 12i - 6 - i \\
 &= 2 + 11i \\
 &= Q \dots
 \end{aligned}$$

$$\therefore \text{Arg}(z^3) = \text{Arg}(X) = \text{Arg}(2+11i) = \angle QOP$$

$$\therefore 3\text{Arg } z = \angle QOP$$

$$\text{Arg } z = \angle XOP$$

$$\therefore \angle XOP = \frac{1}{3} \angle QOP$$

Note: $\frac{1}{3}$ if:

$$\ast \angle QOP = \tan^{-1} \frac{11}{2}$$

and

$$\angle XOP = \tan^{-1} \frac{1}{2}$$

or

$$\ast \angle XOP = \text{Arg}(z) - \text{Arg}(\vec{OQ})$$

and

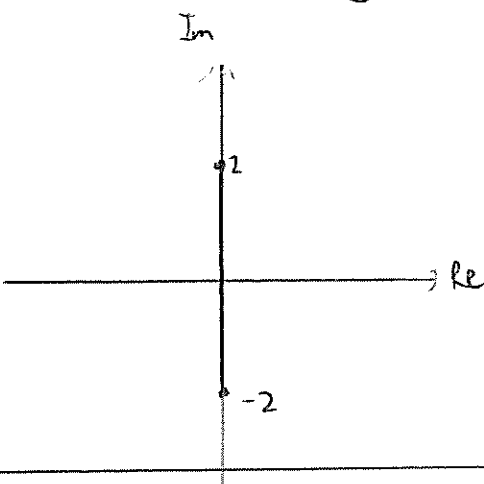
$$\angle QOX = \text{Arg}(\vec{OQ}) - \text{Arg}(z)$$

But not $\frac{2}{3}$ for both!!!

You receive **NO MARKS** for recognising that

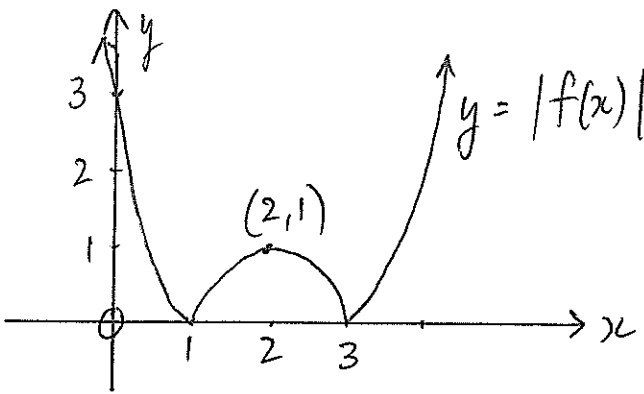
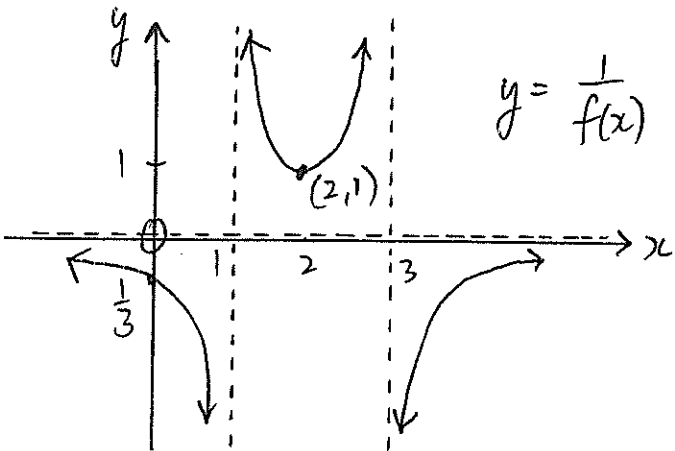
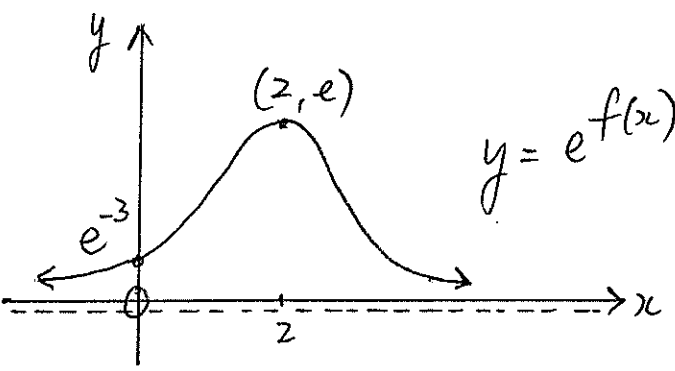
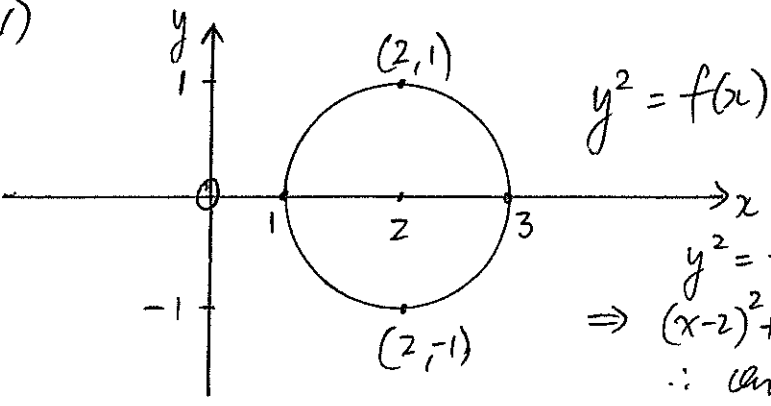
$$|z^3| = |\vec{OQ}| \dots$$

Suggested Solutions	Marks	Marker's Comments
c) i) let $z = x + iy$, $\therefore \bar{z} = x - iy$ $\therefore \frac{z}{\bar{z}} - \frac{\bar{z}}{z} = \frac{x+iy}{x-iy} - \frac{x-iy}{x+iy}$ $= \frac{(x+iy)^2 - (x-iy)^2}{x^2+y^2} \quad \text{--- (1)}$ $= \frac{4xyi}{x^2+y^2} \quad \text{--- (1)}$ which is imaginary. . or $\frac{z}{\bar{z}} - \frac{\bar{z}}{z} = \frac{z^2 - \bar{z}^2}{z\bar{z}} \quad \text{--- (1)}$ $= \frac{(z+\bar{z})(z-\bar{z})}{ r ^2}$ $= \frac{\text{Re}(z) \times \text{Im}(z) i}{ r ^2} \quad \text{--- (1)}$ which is imaginary		

Suggested Solutions	Marks	Marker's Comments
ii) let $z = r(\cos \theta + i \sin \theta)$ $\therefore \frac{4xyi}{x^2+y^2} = \frac{4(r\cos\theta)(r\sin\theta)i}{r^2\cos^2\theta + r^2\sin^2\theta}$ $= \frac{4\cos\theta\sin\theta i}{1}$ $= 2i\sin 2\theta$ $\therefore$ locus is between -2 and 2 on the Imaginary axis.  <u>Note</u> : Students receive 1 for drawing a locus on the Imaginary axis without the restricted domain.	① ①	

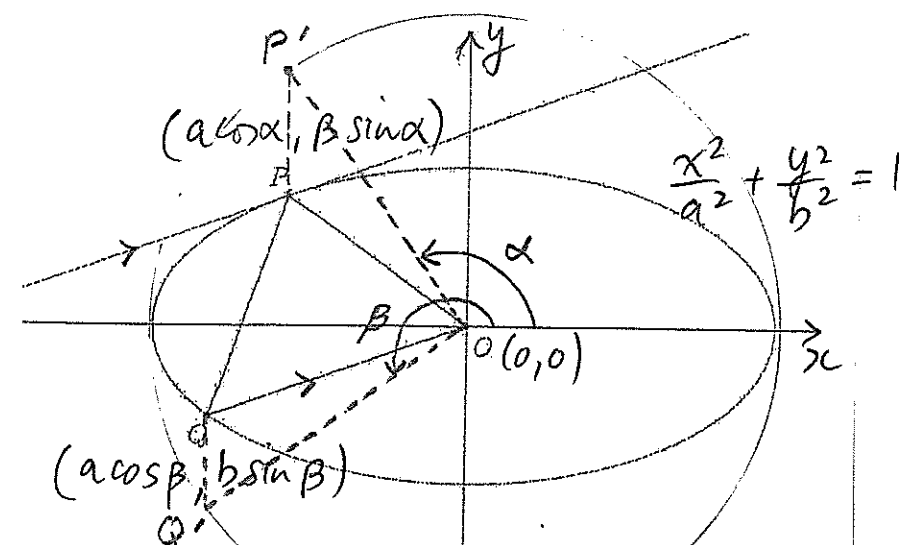
Suggested Solutions	Marks	Marker's Comments
d) i) $P(x) = 0$ (Factor theorem) $\therefore x^4 - 4x^3 + 24x^2 - 40x + 100 = 0$ $x^4 - 4x^3 + 24x^2 - 40x + 100 = 0 \quad \text{--- (1)}$ $\overline{x}^4 - 4\overline{x}^3 + 24\overline{x}^2 - 40\overline{x} + 100 = 0$ $\overline{x}^4 - 4\overline{x}^3 + 24\overline{x}^2 - 40\overline{x} + 100 = 0 \quad \text{--- (1)}$ $\therefore \overline{x}$ is also a root. $\therefore P(\overline{x}) = 0$ (factor theorem) <u>or</u> Polynomial with real coefficients. $\therefore$ If x is a complex root, then $\overline{x}$ is also a root. $\therefore P(\overline{x}) = 0$ (factor theorem)		(1) mark only!!!

Suggested Solutions	Marks	Marker's Comments
ii) let the roots be $\alpha, \alpha, \bar{\alpha}, \bar{\alpha}$, let $\alpha = x + iy$, $\therefore \bar{\alpha} = x - iy$ $\therefore 2\alpha + 2\bar{\alpha} = 4$ (sum of roots) ——— $\therefore 4x = 4$ $x = 1$ ——— $\alpha^2 \bar{\alpha}^2 = 100$ (product of roots) $\alpha \bar{\alpha} = \pm 10$ $\therefore x^2 + y^2 = \pm 10$ Since $x = 1$, $1 + y^2 = \pm 10$ $y^2 = -1 \pm 10$ $y = 9$ only (since $y \in \mathbb{R}$) $\therefore y = \pm 3$ $\therefore$ roots are $1 + 3i$ and $1 - 3i$ ———	① ① ①	

Suggested Solutions		Marks	Marker's Comments
a) (i)		1	correct shape
		1	intercepts + turning points +
(ii)		1	correct shape
		1	asymptotes + intercepts
(iii)		1	correct shape
		1	intercepts, + turning points + asymptotes
(iv)	 $y^2 = f(x)$ $y^2 = -x^2 + 4x - 3$ $\Rightarrow (x-2)^2 + y^2 = 1$ $\therefore$ centre (2, 0) radius 1 unit	1	shape
		1	turning points + vertical tangents

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C (i)



Slope at P: $\frac{dy}{dx} = \frac{dy/d\alpha}{dx/d\alpha} = \frac{b \cos \alpha}{-a \sin \alpha}$

$$= -\frac{b}{a} \cot \alpha$$

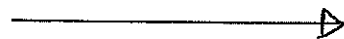
$$m_{OQ} = \frac{b \sin \beta - 0}{a \cos \beta - 0} = \frac{b}{a} \tan \beta$$

But OQ is parallel to tangent at P

$$\Rightarrow m_{OQ} = m(\text{at } P)$$

i.e. $\frac{b}{a} \tan \alpha = -\frac{b}{a} \cot \alpha$

or $\cot \alpha = -\tan \beta$



1

For correct gradient

1

Correctly finding gradient of OQ and equating to gradient at P

Task 2 : 2017

MATHEMATICS Extension 2: Question....9...

JRAHS

Suggested Solutions

Marks

Marker's Comments

$$\begin{aligned} C(ii) \quad |OPQ| &= \frac{1}{2} |x_1 y_2 - y_1 x_2| \\ &= \frac{1}{2} |a \cos \alpha \cdot b \sin \beta - a \cos \beta \cdot b \sin \alpha| \\ &= \frac{1}{2} |ab| |\sin(\beta - \alpha)| \end{aligned}$$

But from (i) $\cot \alpha = -\tan \beta$

$$\Rightarrow \frac{\cos \alpha}{\sin \alpha} = -\frac{\sin \beta}{\cos \beta}$$

$$\Rightarrow \cos \alpha \cos \beta + \sin \alpha \sin \beta = 0$$

$$\Rightarrow \cos(\beta - \alpha) = 0$$

$$\Rightarrow \beta - \alpha = \frac{\pi}{2} \text{ or } -\frac{\pi}{2}$$

Note:

$$\begin{aligned} \cot \alpha &= -\tan \beta \\ &= \tan(-\beta) \end{aligned}$$

$$\Rightarrow \alpha - \beta = \frac{\pi}{2}$$

$$\begin{aligned} \therefore |OPQ| &= \frac{1}{2} |ab| |\pm 1| \\ &= \frac{1}{2} |ab| \end{aligned}$$

$$\begin{aligned} |\sin(\beta - \alpha)| &= \sqrt{\sin^2(\beta - \alpha)} \\ &= \sqrt{1 - \cos^2(\beta - \alpha)} \\ &= \sqrt{1 - 0} \\ &= 1 \end{aligned}$$

1

Correctly for reversing the trig expansion

1

For using (ii) to show $\beta - \alpha = \frac{\pi}{2}$

1

For confirmation that $|\sin(\beta - \alpha)| = 1$

Question 10 1/6

*badly done, handwriting through the entire question was hard to read in

(a)

$$\arg(z_3 - z_1) - \arg(z_2 - z_1) = \pi/3$$

$$\arg(z_1 - z_2) - \arg(z_3 - z_2) = \pi/3$$

$$\arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right) = \arg\left(\frac{z_1 - z_2}{z_3 - z_2}\right) = \pi/3 \dots \dots \textcircled{1}$$

$PQ = QR = PR$ (all sides of an equilateral triangle)

$$\therefore |z_1 - z_2| = |z_3 - z_2| = |z_3 - z_1|$$

$$\text{so } \left| \frac{z_3 - z_1}{z_2 - z_1} \right| = \left| \frac{z_1 - z_2}{z_3 - z_2} \right| \dots \dots \textcircled{2}$$

now if $|z| = |w|$ and $\arg(z) = \arg(w)$ then $z = w$.

$$\text{so } \frac{z_3 - z_1}{z_2 - z_1} = \frac{z_1 - z_2}{z_3 - z_2}$$

$$\therefore (z_3 - z_1)(z_3 - z_2) = (z_1 - z_2)(z_2 - z_1)$$

$$z_3^2 - z_2 z_3 - z_1 z_3 + z_1 z_2 = z_1 z_2 - z_1^2 - z_2^2 + z_1 z_2$$

$$\therefore z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_1 z_3$$

* A lot of students mixed up vectors with modulus, this was a fatal error.

* Students picked up the L.H.S. and suddenly made it equal to zero with no reasons why.

* A lot of basic errors, and then lots of fudging.

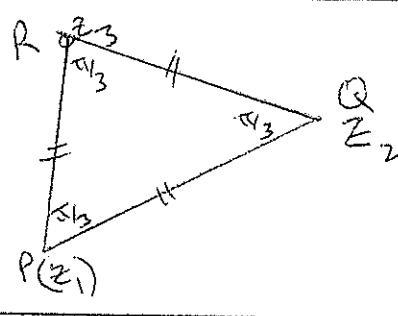
* A clear diagram would have helped more students do better.

or

$$\begin{aligned}\vec{PQ} &= z_2 - z_1 \\ \vec{PR} &= z_3 - z_1 \\ \vec{QR} &= z_3 - z_2\end{aligned}$$

$\pi/6$

(1)



$PQ = PR = QR$ (all sides of an equilateral triangle.)

$$\therefore \frac{PQ}{PR} = \frac{PR}{QR}$$

$$\frac{z_2 - z_1}{z_3 - z_1} = \frac{z_3 - z_1}{z_3 - z_2} \quad (1)$$

$$z_2 z_3 - z_2^2 - z_1 z_3 + z_1 z_2 = z_3^2 - z_3 z_1 - z_1 z_3 + z_1^2$$

$$z_2 z_3 + z_1 z_2 + z_1 z_3 = z_1^2 + z_2^2 + z_3^2 \quad (1)$$

$$\text{ie } z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_1 z_3 + z_2 z_3$$

or

ΔPQR is equilateral (data) $\vec{PQ} = z_2 - z_1$, $\vec{PR} = z_3 - z_1$, $\vec{QR} = z_3 - z_2$
 $\therefore$ All angles are $\pi/3$ and all sides are equal.
 ie $\hat{R}PQ = \hat{P}RQ = \hat{R}QP = \pi/3$

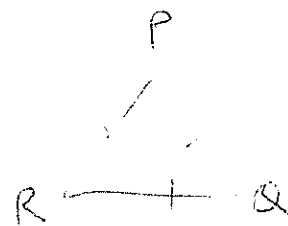
$$\vec{PR} \text{ cis } \pi/3 = \vec{PQ}$$

$$\vec{RQ} \text{ cis } \pi/3 = \vec{RP}$$

$$\vec{PQ} \text{ cis } \pi/3 = \vec{RQ}$$

$$\therefore \text{cis } \pi/3 = \frac{\vec{PQ}}{\vec{PR}}$$

$$\text{cis } \pi/3 = \frac{\vec{RP}}{\vec{RQ}} \quad (1)$$



$$\therefore \frac{z_2 - z_1}{z_3 - z_1} = \frac{z_1 - z_3}{z_1 - z_2}$$

$$(z_2 - z_1)(z_1 - z_2) = (z_1 - z_3)(z_3 - z_1)$$

$$z_2^2 - z_2 z_3 - z_1 z_2 + z_1^2 = z_1 z_3 - z_1^2 - z_3^2 + z_1 z_3$$

$$z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_1 z_3 + z_2 z_3$$

3/6

(b) lower rectangles $< \int_m^{2m} \tan^{-1} x dx < \text{upper rectangles}$

upper rectangles: $1 \times \tan^{-1}(m+1) + 1 \times \tan^{-1}(m+2) + \dots + 1 \times \tan^{-1}(2m)$
 $= \tan^{-1}(m+1) + \tan^{-1}(m+2) + \dots + \tan^{-1}(2m)$
 $= S_m$

$\int_m^{2m} \tan^{-1} x dx =$; let $u = \tan^{-1} x$ $\frac{du}{dx} = \frac{1}{1+x^2}$ $\frac{dv}{dx} = 1$
 $v = x$

$$\begin{aligned} I &= \left[x \tan^{-1} x \right]_m^{2m} - \int_m^{2m} \frac{x}{1+x^2} dx \\ &= 2m \tan^{-1}(2m) - m \tan^{-1}(m) - \frac{1}{2} \int_m^{2m} \frac{2x}{1+x^2} dx \\ &= 2m \tan^{-1}(2m) - m \tan^{-1}(m) - \left[\frac{1}{2} \ln(1+x^2) \right]_m^{2m} \\ &= 2m \tan^{-1}(2m) - m \tan^{-1}(m) - \frac{1}{2} \ln(1+4m^2) + \frac{1}{2} \ln(1+m^2) \\ &= 2m \tan^{-1}(2m) - m \tan^{-1}(m) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right) \end{aligned}$$

$\therefore$ so $\int_m^{2m} \tan^{-1} x dx < S_m$

i.e. $2m \tan^{-1}(2m) - m \tan^{-1}(m) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right)$

now for the lower rectangles:

area of lower rectangles $= 1 \times \tan^{-1}(m) + 1 \times \tan^{-1}(m+1) + \dots + 1 \times \tan^{-1}(2m-1)$
 $= \tan^{-1}(m) + S_m - \tan^{-1}(2m)$

so $\tan^{-1}(m) + S_m - \tan^{-1}(2m) < 2m \tan^{-1}(2m) - m \tan^{-1}(m) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right)$

$$\begin{aligned} S_m &< 2m \tan^{-1}(2m) + \tan^{-1}(2m) - m \tan^{-1}(m) - \tan^{-1}(m) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right) \\ S_m &< (2m+1) \tan^{-1}(2m) - (m+1) \tan^{-1}(m) + \frac{1}{2} \ln \left(\frac{1+m^2}{1+4m^2} \right) \end{aligned}$$

4/6

now, $y = \tan^{-1} x$ is an increasing function
 so $\tan^{-1}(2m+2) > \tan^{-1}(2m)$

$$m > 0, 2m+2 > 0$$

$$\therefore (2m+1)\tan^{-1}(2m) < (2m+2)\tan^{-1}(2m+2)$$

$$(m+1)\tan^{-1}(m) < (m+1)\tan^{-1}(m+1)$$

$$\text{so, } S_m < (2m+1)\tan^{-1}(2m) - (m+1)\tan^{-1}(m) + \frac{1}{2}\ln\left(\frac{1+x^2}{1-x^2}\right)$$

$$\therefore S_m < (2m+2)\tan^{-1}(2m+2) - (m+1)\tan^{-1}(m+1) + \frac{1}{2}\ln\left(\frac{1+x^2}{1-x^2}\right)$$

ii) Lower bound to S_{1000} is:

$$2000\tan^{-1}(2000) - 1000\tan^{-1}(1000) + \frac{1}{2}\ln\left(\frac{1+10^6}{1+4 \times 10^6}\right)$$

$$= 1570.103179 \dots$$

upper bound is:

$$2002\tan^{-1}(2002) - 1001\tan^{-1}(1001) + \frac{1}{2}\ln\left(\frac{1+10^6}{1+4 \times 10^6}\right)$$

$$= 1571.6739$$

$$\therefore S_{1000} \approx 1570, \text{ (3 sig. figures)}$$

* Full marks were awarded if the both bounds were calculated correctly.

* Some students (≈ 50), forgot to put their calculator into RADIANS MODE, so they got:

$$\text{upper limit} = 90.089.30684$$

$$\text{lower limit} = 89.999.30684$$

$$\therefore S_{1000} = 90044$$

$$= 90000 \text{ (3 sig figures)}$$

they were awarded one mark maximum.

(c) (i) $P(z) = z^n + z + p$ — ①

Supposing $P(z) = Q(z)R(z)$ — ②

then by ①, $P(0) = p$

& so by ②

$$P(0) = Q(0)R(0)$$

$$= p$$

Compulsory

As p is prime & Q, R are integer polynomials (i.e. will return integers for integer inputs), then either $Q(0) = \pm 1 \Rightarrow R(0) = \pm p$ or $Q(0) = \pm p$ while $R(0) = \pm 1$. ✓

(ii) If we take, wlog, the case where $Q(0) = \pm 1$, then ± 1 is the constant term of Q .

Hence, for $\alpha_1, \dots, \alpha_k$ the roots of $Q(z) = 0$, we have

$$|\alpha_1 \alpha_2 \dots \alpha_k| = |\pm 1| = 1.$$

*Explanation needed to be clear and convincing ✓

Now, if all the roots were > 1 , then $|\alpha_1 \dots \alpha_k| > 1$ ✗

So at least one root

must be less than or equal to 1.

*A lot of students just wrote the equation out wrong, and hence scored 0 marks ✓

b/p

So there is at least one α_j such that

$$|\alpha_j| \leq 1.$$

(iii). Take $z = \alpha_j$ in $P(z) = Q(z)R(z)$

and in $P(z) = z^n + z + p$. — ①

Then:

$$P(\alpha_j) = Q(\alpha_j)R(\alpha_j)$$

$$= 0 \text{ — ②}$$

So $P(\alpha_j) = \alpha_j^n + \alpha_j + p = 0$ by ① & ②

But then $\alpha_j^n + \alpha_j = -p$

so $|\alpha_j^n + \alpha_j| = |p|$ ✓

But $|\alpha_j^n + \alpha_j| \leq |\alpha_j^n| + |\alpha_j|$

i.e. $|p| \leq |\alpha_j^n| + |\alpha_j|$

Since $|\alpha_j| \leq 1 \rightarrow |\alpha_j^n| \leq 1$

& so $|p| \leq |\alpha_j^n| + |\alpha_j| \leq 1 + 1 = 2$

i.e. $|p| \leq 2$ ✓

But $p > 2$ & prime, so $p \geq 3$. ✓ #

Hence such polynomials cannot be factored as a product of integer polynomials. Therefore, $P(z)$ is irreducible.